School of Computing & IT
Web Programming (CIT2011)
Week 1 – Exploring the Web with Developer Tools

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1. Lab Overview

Modern web developers need to understand what happens inside the browser when a webpage loads and when a user interacts with it. In this lab, students use Chrome, or Firefox Developer Tools to inspect HTML, and CSS.

Register to https://onecompiler.com/. You will use this online compiler to share your ongoing labs with your tutor for review.

2. Learning Outcomes

1. Open and navigate browser Developer Tools.
2. Identify HTML elements using the Elements/Inspector panel.
3. Modify HTML and CSS temporarily using Developer Tools.
4. Use responsive/device emulation to test different screen sizes.
5. Use Developer Tools as part of a modern web-development workflow.

3. Software Requirements

Students need the following:

- Google Chrome / Mozilla Firefox
- Internet connection
- A text editor such as VS Code

4. Developer Tools Overview

Chrome	Firefox	Purpose
Elements	Inspector	Examine HTML and CSS
Console	Console	Execute JavaScript and view errors
Sources	Debugger	Examine JavaScript/source files
Network	Network	Examine HTTP requests
Application	Storage	Examine cookies/storage
Lighthouse	Lighthouse	Performance/accessibility analysis
Device Toolbar	Responsive Design Mode	Test different screen sizes

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PART A — Opening Developer Tools

Activity 1: Open Developer Tools

a. Open https://www.w3schools.com/html/default.asp or another publicly accessible website of your choice).
b. Open Developer Tools using right-click → Inspect, F12, Ctrl+Shift+I on Windows, or Cmd+Option+I on macOS.

Question 1. What is the name of the panel used to inspect the webpage's HTML? Elements

Question 2. List five panels available in Developer Tools.

1. Elements 2. Console 3. Sources

4. Network 5. Performance

PART B — Inspecting HTML

Activity 2: Examine the HTML Structure

Select the Elements/Inspector panel. Locate <html>, <head>, <body>, headings, paragraphs, links, images and navigation elements.

Question 3. Identify one heading element. Record the HTML tag and the text inside it.

HTML tag: <h1>

Text: HTML Tutorial

Question 4. Find one hyperlink. Record the HTML element and its href value.

HTML element: <a>

href: 'javascript:void(0);' class='topnav-icons fa fa-menu w3-hide-large w3-left w3-bar-item w3-button ga-nav' style="line-height:1.1;padding-top:8px!important;padding-bottom:7px!important;" onclick='open_menu()' title='Menu'>

Activity 3: Modify HTML

Find a visible heading. Right-click the element and select Edit as HTML (or double-click it). Temporarily change it to:

<h1>CIT2011 Developer Tools Lab</h1>

Question 5. What happened to the webpage?

Only the heading changed to CIT2011 Developer Tools after editing the HTML.

Question 6. Was the original website permanently changed? Explain why.

No. However the change was temporary because editing HTML in Developer Tools only changes the webpage in the browser. Therefore, refreshing the page restores the original content.

PART C — Inspecting CSS

Activity 4: Examine CSS

Select an HTML element containing visible text. Examine the Styles panel and identify CSS properties controlling font size, font family, margin, padding, text colour and background colour.

Property	Value
font-size	16px
font-family	Arial, sans-serif, Verdana
margin	0
padding	0
color	rgb(0, 0, 0)
background-color	rgb(255, 255, 255)

Activity 5: Modify CSS

Add the following declaration to a heading in the Styles panel:

font-size: 50px;

Then change it to:

font-size: 20px;

Question 1. What happened when the font-size value changed?

The heading got much bigger at 50px, then shrank to 20px, without reloading the page updated instantly.

Question 2. Add text-transform: uppercase; What happened?

All the letters in the heading turned to capitals. The HTML text itself didn't change; only its display did.

Question 3. Change margin and padding. Explain the difference between margin and padding.

Firstly, the padding is the space inside the element, between the content and the border, where it takes the element's background colour. The margin is the space outside the border, which separates the element from other elements

PART D — Understanding the Box Model

Activity 6: Inspect the Box Model

Select an element and locate the Box Model diagram.

Question 4. Identify the four major components of the CSS box model.

1. Content 2. Padding 3. Border 4. Margin

PART I — Responsive Web Design

Activity 15: Device Emulation

Open Device Toolbar / Responsive Design Mode and test mobile, tablet, laptop and desktop sizes such as 375×667, 768×1024 and 1366×768.

Question 5. What happened to the webpage when the screen width became smaller?

The content rearranged to fit the narrower screen. Columns stacked vertically, text wrapped, and the page scrolled more vertically.

Question 6. Did the navigation change? Describe what happened.

It did because the horizontal menu collapsed into a hamburger(=)icon or stacked vertically.

Question 7. Did images resize? Explain your observation.

Yes, the images scaled down to fit the screen width, likely because of max-width:100%. Some images may have been cropped or hidden.

Activity 16: Test Your Own Responsive Design

Create a file named responsive.html using the following code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Responsive Test</title>
  <style>
    body { font-family: Arial; margin: 20px; }
    .container { display: flex; gap: 20px; }
    .box { padding: 30px; background-color: lightblue; flex: 1; }
    @media (max-width: 600px) {
      .container { flex-direction: column; }
    }
  </style>
</head>
<body>
  <h1>Responsive Web Design</h1>
  <div class="container">
    <div class="box">Box 1</div>
    <div class="box">Box 2</div>
    <div class="box">Box 3</div>
  </div>
</body>
</html>
```

Use Developer Tools to test 1200px, 900px, 700px, 600px, 500px and 375px.

Question 8. At approximately what width does the layout change from rows to columns?

600px, the boxes stack into a column at 600px and below.

Question 9. Which CSS feature caused the layout to change?

media query
@media(max-width:600px) → {flex direction: column}

Question 10. Complete the table above.

FINAL CHALLENGE — Activity 20: Become the Web Detective

A student reports that a webpage is displaying incorrectly on their mobile phone, although it works correctly on a laptop. Use Developer Tools to investigate responsive layout, CSS rules, media queries, images, JavaScript errors, network requests and HTTP status codes. 1200px, 900px: row and 600px, 500px, 375px: column

Question 11. Identify three possible reasons why a webpage may work correctly on desktop but incorrectly on mobile.

1. There's no viewport meta tag, or media queries are missing or wrong
2. Fixed widths (e.g. width:1000px) or large images cause overflow and horizontal scrolling.
3. Javascript errors or failed network requests (404) occur on mobile, or the mobile browser doesn't support certain CSS/JS features.

FINAL REFLECTION

Question 12. How does Developer Tools help a web developer understand a webpage?

It shows the page's HTML structure, applied CSS, box model, Javascript errors, and network activity. Which helps a web developer see how the page is built and behaves.

Question 13. Why is it useful to inspect HTML and CSS rather than simply looking at the webpage?

Just looking at a page only shows the result, inspecting shows why it looks that way, meaning which apply and which are overridden. You can also test changes live.

Question 14. Why is the Network panel important to a web developer?

It shows every file the page loads, along with status codes (200, 404, 500), file sizes, and load times. This helps you find missing files and slow resources.

Question 15. Why is responsive testing important?

Users visit on phones, tablets, and desktops. Testing makes sure the site is readable and usable on every screen size.

Question 16. How can Developer Tools help identify bugs?

The console shows Javascript errors with line numbers. Elements/ styles show wrong or overridden CSS. Network shows failed requests.

Question 17. Explain how HTML, CSS and JavaScript work together to create an interactive webpage.

HTML provides the structure and content, CSS controls the appearance and layout, and Javascript adds behaviour.

Question 18 – Build a webpage

Use the template provided by your tutor to build a webpage using HTML5 and basic CSS3. Webpage must include, basic HTML5 tags such as, image, paragraph, navigation, heading, sections, footer, links, etc.

Practical Deliverables

- Completed lab sheet (Questions 1–18).
- At least six screenshots: HTML inspection, CSS inspection
- responsive.html